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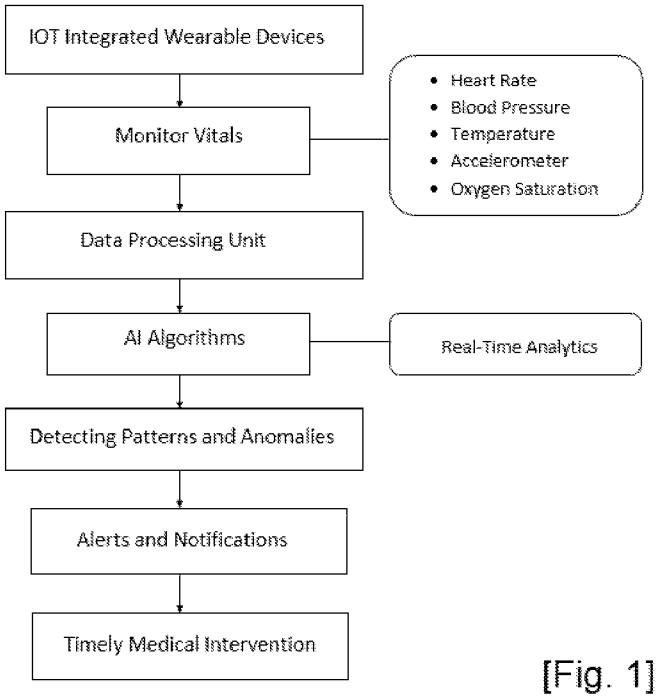
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(54) Title: AI-ENHANCED REMOTE PATIENT MONITORING SYSTEM



[Fig. 1]

(57) Abstract: This invention involves a remote monitoring system that facilitates prompt medical attention and guidance for individuals, eliminating the necessity for frequent hospital visits. Individuals who are not in a critical state or find hospital costs burdensome can choose to recuperate at home while being remotely supervised by artificial intelligence and relevant medical professionals. Prescriptions are issued as needed. Additionally, it enables individuals to have healthcare management that is tailored to their own needs and is more convenient. This application employs a smartphone camera and a blood pressure and pulse measurement device for its intended purposes. This novel technique not only reduces the amount of time and money spent by the patient, but also enables a more effective use of healthcare resources. Patients can access prompt medical care and advice without the necessity of repeated visits to a healthcare center.

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[Continued on next page]

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Description

Title of Invention: AI-Enhanced Remote Patient Monitoring System

[0001] Individuals who are not in critical condition or those who are unable to afford hospital costs can find It difficult to benefit from proper healthcare. This claimed invention addresses this problem and introduces a remote monitoring system designed to provide quick medical attention and guidance for individuals, thereby reducing the need for frequent visits to the hospital. Patients who are not in critical condition or those who are burdened by the costs of hospital visits can now recover at home with the help of artificial intelligence (AI) and remote monitoring by medical professionals. If required, medical prescriptions can be issued remotely through this system.

[0002] Furthermore, this system enables individuals to easily manage healthcare services tailored to their specific needs from the comfort of their homes. The mobile application utilizes the camera on smartphones, as well as measures blood pressure and heart rate for personalized monitoring purposes. This state-of-the-art technology not only saves patients time and money but also ensures a more efficient allocation of medical resources. Using this system,patients can receive timely medical attention and guidance without the hassle of making multiple visits to the healthcare facility.

Technical Field

[0003] G16H – A61B5

Background Art

[0004] IN202321040143

[0005] REMOTE MONITORING OF HIGH-RISK PATIENTS USING ARTIFICIAL INTELLIGENCE

[0006] A remote monitoring system utilizing artificial intelligence for high-risk patients is disclosed. The system comprises data collection devices, a communication network, a data processing unit, an AI engine, and a user interface. Patient data is collected in real-time, transmitted securely, and analyzed using advanced AI techniques. The system provides early identification of health risks, personalized insights, remote accessibility, improved efficiency, and enhanced patient care.

[0007] This mentioned patent resembles our claimed system in its function. However, this invention uses AI for data processing and analytics while ours does not rely on It because of the high rate of inaccuracy. Moreover, our system has a primary focus on providing a cost-effective solution to help reduce medical expenses in the long run while this patent makes no mention of this issue. Our designed system can also issue prescriptions whenever needed, whereas this system lacks this feature.

[0008] US20220400989

[0009] TREATMENT AND DIAGNOSES OF DISEASE AND MALADIES USING REMOTE MONITORING, DATA ANALYTICS, AND THERAPIES

[0010] An example system for enhanced remote monitoring of a patient can obtain measurements of the patient's vital signs using various computer devices, such as wearable devices (e.g., smartwatch) and mobile devices (e.g., smartphone). The system implements a video-based vital sign capture function, which operates the mobile device as an optic sensor in order to obtain vital sign measurements for the patient from video imaging data of the body of the patient. The video imaging data is captured using a digital camera of the mobile device. The system can include a diagnostics server connected over a network to a data server that may host medical data silos, which is further connected to AI machine learning systems.

[0011] Even though this mentioned system bears a strong resemblance to our claimed invention in both function and applicability, it focuses primarily on the psychological aspect of the patients medical needs and employs the iCBT method. Our design, on the other hand offers medical advice and services on all aspects and is not limited to one method. Moreover, our claimed system can issue prescriptions when needed, while this system makes no mention of this feature. Also, this system uses a digital camera for obtaining measurements while ours relies on a safer option that is IOT.

[0012] US20220047160

[0013] CEILING AI HEALTH MONITORING APPARATUS AND REMOTE MEDICAL-DIAGNOSIS METHOD USING THE SAME

[0014] A ceiling artificial intelligence (AI) health monitoring system, includes a monitoring device provided on a ceiling of a space to acquire health information from a patient, and a medical management device configured to apply the health information of the patient acquired to an artificial intelligence-based learning model to determine health condition information of the patient and provide the determination result to a doctor monitor. The medical management device is configured to provide remote medical diagnosis information from the doctor monitor to a user terminal.

[0015] This mentioned invention share some common ground with our designed patient monitoring system in their overall function. However the method is completely different since this patent involves a contraption installed on the ceiling of a space and uses sensors and thermographic images while ours uses a plurality of devices that can be worn by the patient. Also this one provides a rather limited range of services for example it does not issue prescriptions if needed and does not include a user friendly interface.

[0016] IN202311045889

[0017] MONITORING HEALTH USING ARTIFICIAL INTELLIGENCE

[0018] The invention proposes a system and method for an AI-based health tracker that utilizes artificial intelligence to analyze and monitor an individual's health, providing personalized recommendations and insights for improving well-being. By processing and analyzing large amounts of health data, the AI algorithms can identify patterns and trends that may be overlooked by humans, aiding in the diagnosis, treatment, and prevention of diseases. The system comprises sensor modules for tracking vital signs, a microprocessor running machine learning algorithms, a long-lasting battery, and connectivity modules for data syncing and transfer. The collected data is preprocessed, analyzed, and stored in a database, allowing the health tracker to offer tailored feedback and suggestions based on the user's unique health profile. Through the integration of AI, this technology enhances health monitoring capabilities, empowering individuals to take proactive steps towards their health and leading to a more personalized and effective approach to healthcare.

[0019] This mentioned invention offers health monitoring and uses artificial intelligence similar to our claimed patent. However, the function of this device is more that of a health tracker assissant, while our system provides detailed medical care options using a plurality of wearable devices.

[0020] CN104873180

[0021] WIRELESS REMOTE MULTIFUNCTIONAL PATIENT MONITOR SYSTEM

[0022] A wireless remote multifunctional patient monitor system comprises a monitoring center, wireless communication units and multifunctional patient monitors, wherein the monitoring center comprises a monitor console, a data backup server, a data analysis server and a data receiving device, and the data receiving device is connected with the monitor console, the data backup server and the data analysis server respectively; each wireless communication unit comprises a data sending device and a data relay device, the data sending device is in wireless data connection with the data receiving device of the monitoring center, and the data relay device is in wireless data connection with the multifunctional patient monitors; a remote monitoring PDA (personal digital assistant) is connected with the data receiving device through a 3G wireless communication network. According to the wireless remote multifunctional patient monitor system, patient sign information acquired by the multifunctional patient monitors can be transmitted to the monitoring center through wireless communication, a doctor can be connected to the monitoring center through the remote monitoring PDA at any time, so that the doctor can acquire sign data of a patient in real time.

[0023] This mentioned invention performs wireless patient monitoring mechanisms with a similar function as our claimed design but the procedure and technology are different, for instance, our system utilizes artificial intelligence and IOT and features a user friendly interface while this one does not.

[0024] CA2997552C

[0025] Integrated medical device and home based system to measure and report vital patient physiological data via telemedicine

[0026] An integrated home based system to measure and report vital patient physiological data via telemedicine is disclosed. The integrated medical device is a personal, affordable, portable medical monitor, providing multiple critical vital sign data for real-time face-to-face communication with qualified health care professionals, direct from the comfort of your home (or wherever you may be travelling), whenever you need it. It is also linked to a secure patient medical record so the patient and/or healthcare professional can collect, archive and track information and trends.

[0027] This invention involves involves telehealth services similar to ours claimed system, however, its function is based on a specific device that is useful as an examination solution, while ours provides a wider range of medical services and uses artificial intelligence as well as several wearable devices for achieving optimal accuracy.

[0028] US20070073266A1

[0029] Compact wireless biometric monitoring and real time processing system

[0030] Systems and methodologies that regulate in real time biometric indicia of an ambulatory patient via employing a distributed computing arrangement of modular component(s), which are tailored in part based on requirements of data to be measured and/or administered. Accordingly, the system can be scaled for different biometric requirements (e.g., data bits, operating frequencies and the like). Such an arrangement can regulate drug delivery units and/or acquire biometric data from an ambulatory patient.

[0031] This invention resembles the function of our claimed system in their use of artificial intelligence but the method and the componants are different for instance this system includes a belt and a memory storage device while ours relies on multiple wearable devices. Also, this system emphasizes drug delivery and biometric data acquisition while ours delivers several other telemedical services as well.

Summary of Invention

[0032] This patent presents a suite of wearable devices embedded with advanced sensors to monitor vital signs such as heart rate, blood pressure, activity levels, and more. These devices seamlessly integrate with a cloud-based platform, powered by AI algorithms, to analyze the collected data in real-time and provide personalized insights and recommendations to users. Through a user-friendly mobile application and web interface, users can monitor their health metrics, set personalized goals, receive alerts for abnormal readings, and collaborate with healthcare providers for proactive care management. This invention addresses the healthcare needs of individuals who

struggle to monitor their health effectively due to limited access to real-time health data, fragmented healthcare systems, and the lack of personalized insights.

Technical Problem

[0033] Accessing healthcare facilities and enduring lengthy appointment wait times can result in significant delays, often spanning several months, in numerous nations. This unfortunate situation can potentially jeopardize an individual's well-being and even lead to loss of life. The potential ramifications of this delay in accessing medical treatment might be severe, particularly for individuals with pressing healthcare requirements. Healthcare systems must prioritize the implementation of strategies to decrease waiting times and enhance timely access to treatment. Furthermore, it reduces expenses. Telemedicine provides a solution by enabling patients to remotely confer with healthcare providers, hence decreasing the necessity for in-person visits. This not only reduces the amount of time patients spend but also enhances the allocation of resources within the healthcare system. Through the utilization of telemedicine, patients can promptly receive medical attention, bypassing the generally lengthy waiting periods commonly associated with traditional healthcare facilities. This not only enhances efficiency and cost-effectiveness for patients, but also improves the availability of healthcare services for individuals residing in distant or underserved regions.

Solution of Problem

[0034] The development of this technology stems from recognizing the challenges individuals face when accessing timely and affordable healthcare. Long wait times in hospitals and the financial burden associated with frequent visits prompted the need for a more accessible and cost-effective solution. The idea is to allow individuals who are not in critical condition or cannot afford hospital expenses to recuperate at home while still receiving proper medical supervision. This involves leveraging artificial intelligence and the expertise of medical professionals to remotely monitor patients and provide necessary medical guidance.

[0035] The solution involves the development of a mobile application that utilizes a smartphone camera along with additional medical devices such as a blood pressure monitor and pulse measurement device. These tools enable the remote monitoring of vital signs and other health indicators.

[0036] Once the app is installed on the patient's smartphone, they can use it to capture vital signs such as blood pressure and pulse. These data points are then transmitted securely to a central platform where artificial intelligence algorithms analyze them in real-time. Relevant medical professionals, such as doctors or nurses, have access to the platform

and can review the patient's data remotely. They can intervene as necessary, issuing prescriptions or providing medical advice through the app's communication features.

[0037] The processing unit in the wearable device typically consists of a microcontroller or a microprocessor. These are specialized integrated circuits designed to execute instructions and process data. The microcontroller or microprocessor runs algorithms for real-time analysis of the health data collected by the sensors. These algorithms may include signal processing techniques, statistical analysis, machine learning models, or a combination thereof. The processing unit filters and preprocesses the raw sensor data to remove noise, artifacts, and irrelevant information. This ensures that only relevant and accurate data are used for analysis. It also extracts features from the preprocessed data that are indicative of the user health status. These features may include heart rate variability, blood pressure trends, activity levels, sleep patterns, and other physiological parameters.

[0038] The microcontroller or microprocessor executes the real-time analysis algorithms using the extracted features as input. These algorithms analyze the data to detect patterns, anomalies, trends, or critical events such as irregular heartbeats or sudden changes in vital signs. Based on the results of the analysis, the processing unit may make decisions or trigger actions such as generating alerts, sending notifications to the user or healthcare provider, adjusting device settings, or initiating interventions.

[0039] The processing unit is designed to operate efficiently in terms of power consumption and optimizes Wi-Fi and Bluetooth connectivity, IOT data transmission, LPWAN technology and integrates with cloud services. It also Implements encryption, authentication, and access control mechanisms that helps protect sensitive health data from unauthorized access or interception during transmission over the network.

[0040] By integrating the wearable device with the appropriate connectivity module and leveraging IoT technologies, healthcare providers can access real-time health data remotely, monitor patients' vital signs continuously, and intervene promptly in case of any anomalies or emergencies, thereby improving patient outcomes and enhancing overall healthcare delivery.

Advantage Effects of the Invention

[0041] Access to medical supervision at home.

[0042] Convenience in capturing and transmitting vital signs and health data.

[0043] Timely Intervention of healthcare professionals.

[0044] Personalized Care.

[0045] Cost-Effectiveness by reducing healthcare expenditures in the long run.

[0046] Data analysis and evidence-based healthcare decisions.

[0047] Improved Patient Experience.

[0048] Scalability and Flexibility.

Brief Description of Drawings

[0049] [Fig.1] Shows a general flowchart of the system's process.

Description of Embodiments

[0050] [Fig.1] Shows a flowchart of the system's mechanism. The hardware includes multiple devices that monitor a variety of health parameters such as heart rate and blood pressure. The devices are integrated with a central data processing unit where Artificial Intelligence algorithms help provide real-time analysis. The data is stored and examined to detect any patterns and anomalies and generate alerts or notifications in case of critical events. This process allows for timely medical intervention by health-care professionals who will communicate with the patient through the user-friendly interface and offer advice or prescriptions.

Examples

[0051] The patient wears the multiple devices and through the internet of things the collected data will be stored and processed. The data is shared with the patient and doctors or other medical care-givers through the mobile application and software. There, they can view and track information such as blood pressure, heart rate, temperature, etc. and run analytics through artificial intelligence and the related medical history. This allows for monitoring all the vital data regarding the patient remotely and safely.

[0052] The wearable devices that collect data on the patient's vital health parameters may include:

[0053] Heart Rate Sensor: Measures the heart rate continuously.

[0054] Blood Pressure Monitor: Monitors blood pressure fluctuations.

[0055] Temperature Sensor: Monitors body temperature changes.

[0056] Accelerometer/Gyroscope: Detects movement and activity levels.

[0057] Oxygen Saturation Sensor (Pulse Oximeter): Measures the oxygen saturation level in the blood.

Industrial Applicability

[0058] This invention will be very widely used in hospitals, clinics, and healthcare facilities can integrate this technology into their existing systems to offer remote monitoring services to patients. This not only enhances patient care but also improves operational efficiency by reducing the need for in-person consultations and hospital admissions.

[0059] Companies specializing in healthcare technology can develop and market platforms and applications tailored for remote patient monitoring. Manufacturers of medical devices such as blood pressure monitors, pulse oximeters, and other vital sign monitoring equipment can capitalize on the growing demand for connected devices.

[0060] It can also be beneficial for Telemedicine platforms to incorporate remote patient monitoring and improve their virtual care services. Health insurance companies may incentivize the adoption of remote monitoring technology among their policyholders. Additionally, continued innovation in remote patient monitoring technology opens up opportunities for research and development in areas such as artificial intelligence, data analytics, and wearable technology. Companies and academic institutions can collaborate to further refine these technologies and explore new applications for remote monitoring in healthcare.

Claims

[Claim 1] An AI Enhanced Remote Patient Monitoring system is designed that contains a plurality of wearable devices, a data processing unit and a software application.

[Claim 2] According to claim 1, the system utilizes data from multiple IOT integrated devices and sensors that monitor the patient's vital signs including heart rate, blood pressure, temperature, accelerometer and oxygen saturation.

[Claim 3] According to claim 1, the devices are integrated with a data processing unit that includes a microprocessor or microcontroller and runs algorithms, filters irrelevant data and offers real-time data analysis.

[Claim 4] According to claim 1, the data processing platform uses artificial intelligence to run algorithms in order to detect patterns, anomalies, trends, or critical events

[Claim 5] According to claim 1, the system provides customizable notifications and alerts based on data analysis for the health care provider to intervene, offer medical advice or issue prescriptions.

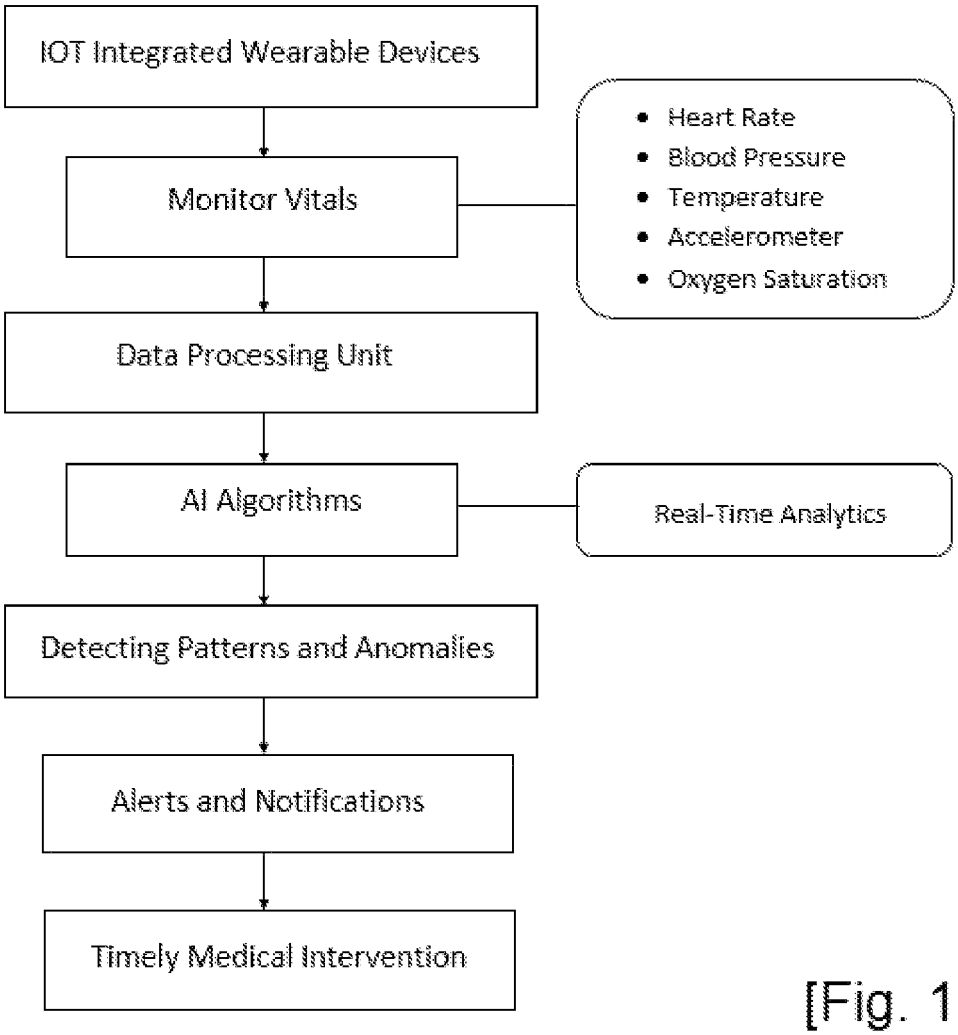
[Claim 6] According to claim 1, the system is designed using optimization techniques in order to minimize power consumption and be cost-effective.

[Claim 7] According to claim 1, the system features modules for communication, connectivity and cloud services, so the data transmission can be facilitated using Wi-Fi, Bluetooth, 4G LTE, 5G and NB-IOT.

[Claim 8] According to claim 1, the software ensures secure data transmission by employing data encryption, authentication, and access control mechanisms.

[Claim 9] According to claim 1, the software can be installed on a personal smartphone and features a user friendly interface that can provide communication with health-care providers based on the patient's health indicators.

[Fig. 1]


IOT Integrated Wearable Devices
Monitor Vitals
Heart Rate
Blood Pressure
Temperature
Accelerometer
Oxygen Saturation
Data Processing Unit
AI Algorithms
Real-Time Analytics
Detecting Patterns and Anomalies
Alerts and Notifications
Timely Medical Intervention

[Fig. 1]

INTERNATIONAL SEARCH REPORT

International application No.
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A. CLASSIFICATION OF SUBJECT MATTER
A61B5/00,G16H10/60,G16H40/67 Version=2024.01

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B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

A61B,G16H

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic database consulted during the international search (name of database and, where practicable, search terms used)

PatSeer, IPO Internal Database

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	Jie Wan, et al. "Wearable IoT enabled real-time health monitoring system" J Wireless Com Network 2018, 298 (2018) 22 December 2018 (22-12-2018) URL: https://doi.org/10.1186/s13638-018-1308-x Whole Document	1-9
X	US20210151179A1 (RAJLAKSHMI DIBYAJYOTI BORTHAKUR)20, May 2021 (20.05.2021) Whole Document	1-9

☐ Further documents are listed in the continuation of Box C. ☒ See patent family annex.

* Special categories of cited documents:
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INTERNATIONAL SEARCH REPORT
Information on patent family members

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Citation	Pub.Date	Family	Pub.Date
US 2021151179 A1	20-05-2021	WO 2019026092 A1	07-02-2019
		CN 111356993 A	30-06-2020
		IN 201741004103 A	08-02-2019